

LAB 21/22 RESULTS

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OVERVIEW

Objective

- Design, build, and test an instrument droid utilizing a breadboard and an Arduino
- Test multiple voltage sources by characterizing Thevenin voltage, output resistance, and current response parameters
- Utilize the Arduino alongside an ADC and DAC and write code to record precise measurements to be analyzed

Measurements

- Record at least 20 measurements for current (mA), Thevenin voltage, Thevenin resistance, and the loaded voltage
- Multiple different input sources should be tested (waveform generator, 5V, 9V supply)

SETUP

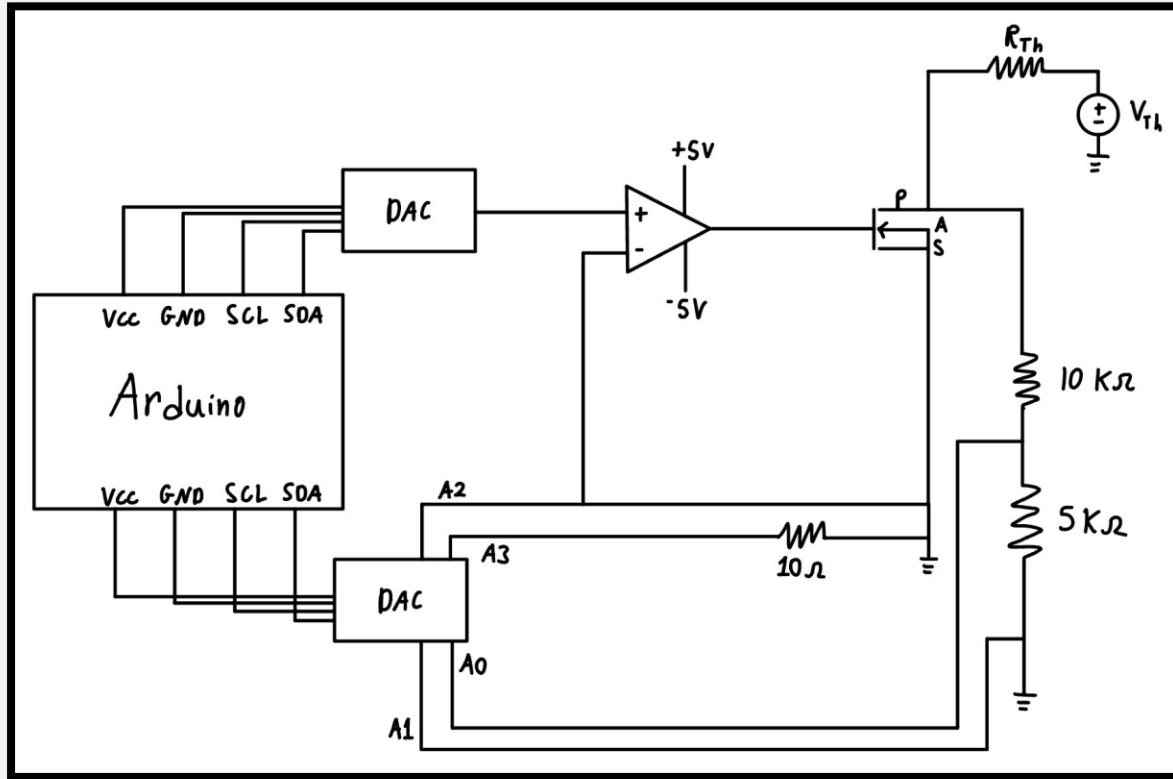


Figure 1: Napkin Sketch

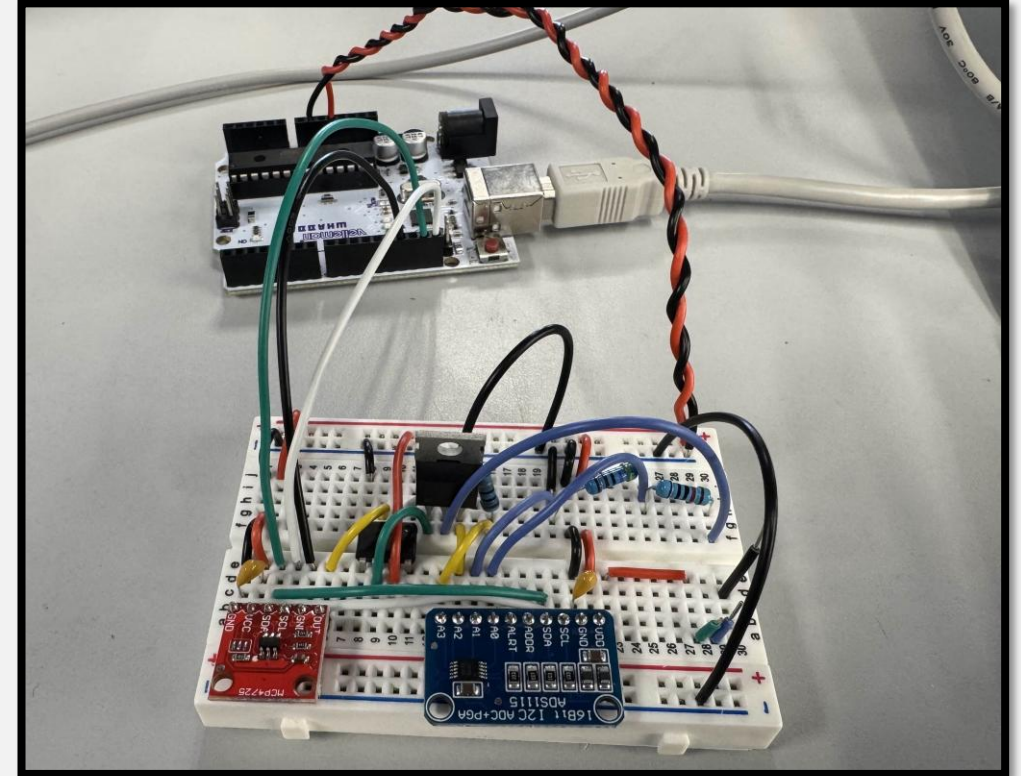


Figure 2: Assembled Circuit

OBSERVATIONS

The load is set by forcing the source to a known voltage using a DAC and an OPAMP. V_{Thev} and R_{Thev} is calculated by measuring the drain voltage of the MOSFET through a feedback path and ADC. The source pin is forced to 0V and by measuring the drain voltage we find V_{Thev} . As the load current increases, R_{Thev} can be found. This was done for each of the three sources.

DATA

Three sources are measured: An Arduino 5V supply, a function generator's 5V DC output, and a 6V 2A power supply. The Thevenin Resistance vs Current is plotted.

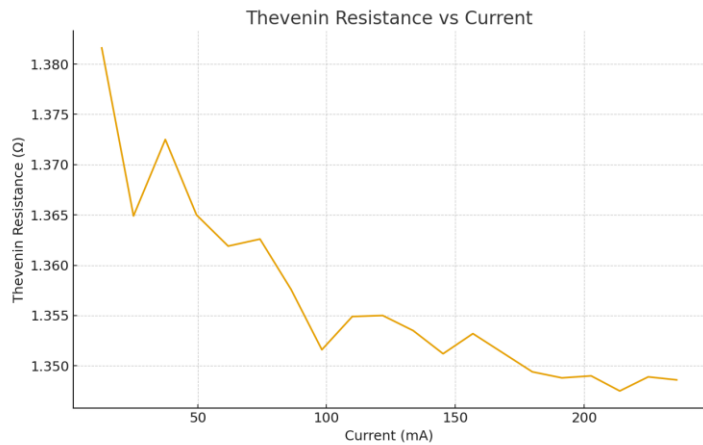


Figure 5: Arduino 5V
 $R_{\text{Thev}}: \sim 1.35 \Omega$

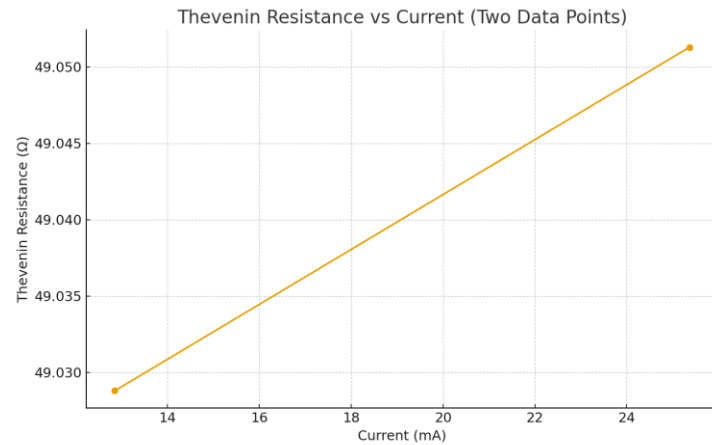


Figure 6: 5V DC Function Generator
 $R_{\text{Thev}}: \sim 50 \Omega$

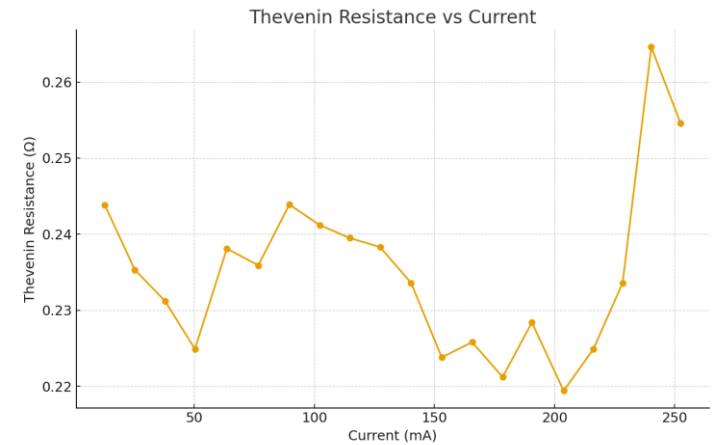


Figure 7: 6V Power Supply
 $R_{\text{Thev}}: \sim 0.25 \Omega$

DAC AND OPAMP OUTPUTS (ARDUINO 5V)

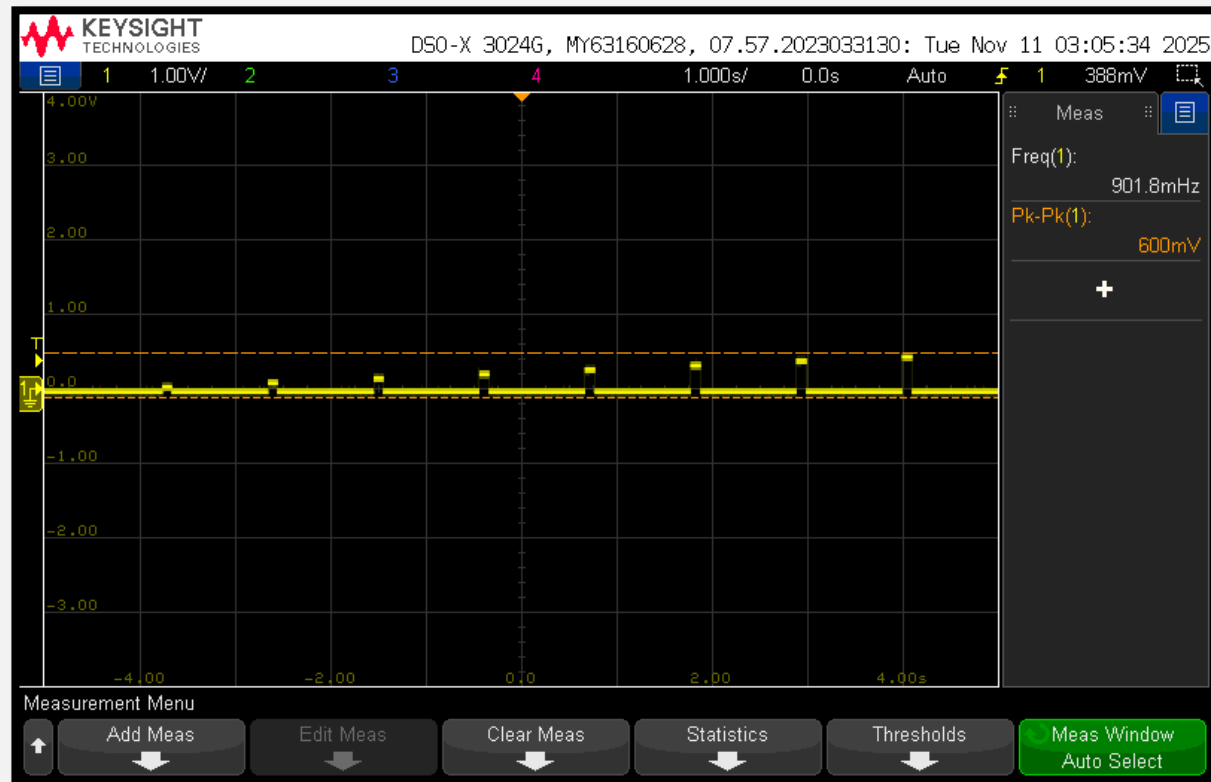


Figure 3: DAC Output

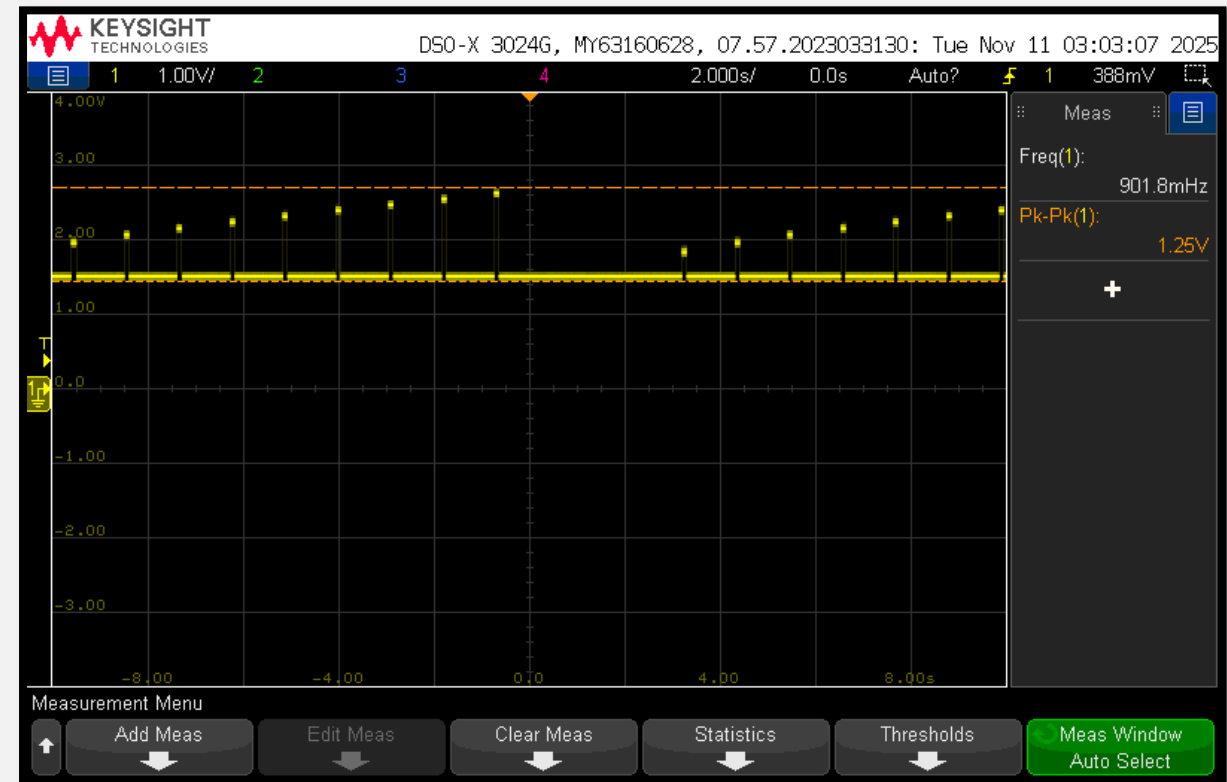


Figure 4: OpAmp Output

CONCLUSION

An instrument droid is successfully assembled to measure various parameters of multiple different voltage source inputs. The data is recorded and plotted on easy-to-read R_{Th} vs Current graphs.

Comparing measurements to expected values successfully validates the data and allows for better understanding of the behaviors of different supply sources.